🔑 **Answer Key — Teacher Copy · Lesson 8-3**

**MSTAR PRACTICE WORKSHEET**

# Lesson 8-3: Write and Solve Equations Using Multiplication or Division

Grade 6 Mathematics · Standard 6.AT.8

*These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.*

## Part A — Skills Check

One point each.

**Question 1** SELECTED RESPONSE — CORRECT: A

Solve: $6x = 42$

- (A) $x = 7$
- (B) $x = 36$
- (C) $x = 48$
- (D) $x = 252$

Divide both sides by 6: $x = 42 \div 6 = 7$. Check: $6 \times 7 = 42$ ✓

**Question 2** SELECTED RESPONSE — CORRECT: D

Solve: $n / 3 = 8$

- (A) $n = 3$
- (B) $n = 5$
- (C) $n = 11$
- (D) $n = 24$

Multiply both sides by 3: $n = 8 \times 3 = 24$. Check: $24 / 3 = 8$ ✓

**Question 3** SELECTED RESPONSE — CORRECT: A

Solve: $9m = 63$

- A $m = 7$
- B $m = 54$
- C $m = 72$
- D $m = 567$

> Divide both sides by 9: $m = 63 \div 9 = 7$. Check: $9 \times 7 = 63$ ✓

**Question 4** SELECTED RESPONSE — CORRECT: D

Solve: $y / 7 = 5$

- A $y = 2$
- B $y = 7$
- C $y = 12$
- D $y = 35$

> Multiply both sides by 7: $y = 5 \times 7 = 35$. Check: $35 / 7 = 5$ ✓

**Question 5** SELECTED RESPONSE — CORRECT: A

A detective splits 54 clues equally among 9 case files. The equation $9c = 54$ represents this. What is c?

- A $c = 6$
- B $c = 9$
- C $c = 45$
- D $c = 63$

> Divide both sides by 9: $c = 54 \div 9 = 6$ clues per file.

**Question 6** SELECTED RESPONSE — CORRECT: A

Vault A is sealed with $4x = 32$. Crack the lock: what is x?

- **A** $x = 8$
- **B** $x = 28$
- **C** $x = 36$
- **D** $x = 128$

> The 4 is multiplying x, so divide both sides by 4: $x = 32 / 4 = 8$. Check: $4 \times 8 = 32$.

## Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

**Question 7** TWO-PART QUESTION

**Part A — correct answer: A**

What is the solution of $6m = 54$?

- **A** $m = 9$
- **B** $m = 48$
- **C** $m = 60$
- **D** $m = 324$

> In 6m the coefficient 6 is multiplied by the variable, so divide both sides by 6: $m = 54 \div 6 = 9$. Check by substituting: $6 \times 9 = 54$.

- **B:** That subtracts 6 from 54. In 6m the 6 multiplies m, so divide instead.
- **C:** That adds 6 to 54. The 6 multiplies m, so divide both sides by it.
- **D:** That multiplies 6 by 54. To undo a multiplication, divide.

**Part B — correct answer: A**

Which reasoning justifies the step used to find that solution?

- **A** Division undoes multiplication, so dividing both sides by 6 isolates the variable.
- **B** Subtracting 6 from both sides removes the 6 that is written next to the variable.
- **C** Multiplying both sides by 6 isolates the variable because the same operation must be repeated.
- **D** The coefficient can simply be dropped because 6 is smaller than 54.

> In 6m the variable is multiplied by 6, so the inverse operation is division. Dividing both sides by 6 keeps the equation balanced and gives $m = 9$. Subtracting 6 would give 48, and $6 \times 48$ is not 54.

**Question 8** SELECT ALL THAT APPLY — CORRECT: A, B, E

Select ALL equations that have $x = 8$ as their solution.

- A $4x = 32$
- B $x \div 2 = 4$
- C $3x = 21$
- D $x \div 4 = 32$
- E $56 = 7x$

Substitute 8 for the variable in each equation. $4 \times 8 = 32$ is true, $8 \div 2 = 4$ is true, and $56 = 7 \times 8$ is true. The third is false because $3 \times 8 = 24$, not 21; its solution is $x = 7$. The fourth is false because $8 \div 4 = 2$, not 32; its solution is $x = 128$.

## Part C — Show Your Thinking

Model answer and scoring guidance.

**Question 9** WRITTEN RESPONSE · 2 POINTS

A bakery packed muffins into boxes of 8. They made 96 muffins total. Write an equation, solve it, and explain why your answer is reasonable. Then write a different situation that uses division and has the same answer.

**Model answer:** Eight muffins per box times an unknown number of boxes equals 96, so the equation is $8x = 96$. Dividing both sides by 8 gives $x = 12$ boxes, and it is reasonable because 12 boxes × 8 muffins rebuilds all 96. A division situation with the same answer works too, such as 60 stickers shared equally by 5 students.

*Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.*

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